

Making Decisions

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These slides are adapted from materials developed by Dr Mark van der Wilk.

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Taking Action

We have been discussing how to build a posterior distribution given observations...

What do you want to **use** a posterior for?

- Posterior is a belief over what will happen.
- You use beliefs to take action / make decisions.
- Uncertainty helps you to balance risk, reward, and new information.

We *learn* about the world,
so we can *act* in it,
to get outcomes that we *desire*.

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As with uncertain reasoning, we can use maths to tell us how to make rational decisions, if we write down our wishlist of axioms.

Paraphrased axioms:

1. Comparing probability distributions A and B over outcomes, you either prefer one over the other, or you have no preference.
2. If you prefer A over B, and B over C, then you prefer A over C.
3. Reduction of compound lotteries: If you prefer A over B, then you also prefer a 50% chance of getting A over a 50% chance of getting B, with a 50% chance of getting C.

See Ch 16 "Artificial Intelligence: A Modern Approach", Russell (2010)

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Von Neumann-Morgenstern Utility Theorem

- If you behave according to these axioms, then there exists a **utility function** that you maximise.
- In other words, if you believe you want to act according to those axioms, you should follow the principle of **Maximum Expected Utility**.

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Principle of Maximum Expected Utility

1. Define a utility function $U : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}$
 $\mathcal{X}$: Space of outcomes. $\mathcal{A}$: Space of actions. Quantifies how good an outcome is, if you take a particular action.
2. Compute expected utility for your actions.
Your beliefs are a distribution over outcomes given action.

$$u(a) = \mathbb{E}_{p(x|\mathcal{D},a)} [U(x, a)] \quad (1)$$

3. At the time of decision making, choose action which maximises expected utility.

$$a^* = \operatorname{argmax}_a u(a) \quad (2)$$

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Exercise: Why do we predict the mean?

Your regression model gives you $p(y|X, \mathbf{y})$. You need to give a “best guess” y_p , and your utility will be $U(y, y_p) = -(y - y_p)^2$.

→ Find y_p in terms of properties of $p(y|X, \mathbf{y})$.

Solution: We want $\frac{d}{dy_p} u(y_p) = \frac{d}{dy_p} \mathbb{E}_{p(y|X, \mathbf{y})} [U(y, y_p)] = 0$.

$$\begin{aligned} \frac{d}{dy_p} u(y_p) &= \frac{d}{dy_p} \int_{-\infty}^{\infty} p(y|X, \mathbf{y}) U(y, y_p) dy \\ &= - \int p(y|X, \mathbf{y}) \frac{d}{dy_p} (y - y_p)^2 dy \\ &= 2 \int p(y|X, \mathbf{y}) (y - y_p) dy \end{aligned}$$

Leibniz Integral Rule

$$0 = 2 \mathbb{E}_{p(y|X, \mathbf{y})} [y] - 2y_p$$

$$y_p = \mathbb{E}_{p(y|X, \mathbf{y})} [y] , \quad \text{i.e. } y_p \text{ should be the mean!}$$

Bonus: what should we predict if utility is absolute error?

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- Maximum Expected Utility: Principle for taking an action based on beliefs.
- SDM: How do we plan for decisions made one after each other?
- Key idea: At each point, apply MEU, given all the knowledge you have at that point
- Consequence: After making an earlier decision, you can **learn**. This information has value.
- This is key reason why things become complicated, but principle is no more complex!

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Example: Commuting (I)

Paula has an exam at 9am today, and it is 8:35am. She will not be admitted if she is late, so her loss function (negative utility) is

$$L(t, a) = \begin{cases} 0 & \text{if } t \leq 25 \text{ min} \\ 100 & \text{if } t > 25 \text{ min} \end{cases} . \quad (3)$$

Paula decides to take a taxi. The taxi driver knows of routes A and B. Route A can either be clear and fast ($Q = 0$) or jammed and slow ($Q = 1$). The time for route B follows a Gaussian.

$$t_A = \begin{cases} 10 \text{ min} & \text{if } Q = 0 \\ 50 \text{ min} & \text{if } Q = 1 \end{cases} , \quad P(Q) = \begin{cases} 0.5 & \text{for } Q = 0 \\ 0.5 & \text{for } Q = 1 \end{cases} , \quad (4)$$

$$p(t_B) = \mathcal{N}(t_B; 25, 5^2) . \quad (5)$$

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Example: Commuting (II)

The driver suggests the following strategy:

- Drive the first 2.5 minutes of route A to see whether it is clear (i.e. to observe Q).
- At that point, decide whether to keep going with route A, or to drive back 2.5 minutes and take route B.

Question: Should Paula:

- Ask the driver to take route A?
- Ask the driver to take route B?
- Ask the driver to continue with his strategy?

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Decision Trees

- Arrange sequence of observations and decisions into a **tree**
- Helps us to compute utility based on decisions
- Helps us to find sequence of optimal decisions

- Rectangle: Decision nodes.
- Circle: Observation of a random variable.
- Leaf nodes: Outcome of utility.
- Nodes are functions of the outcome of parent RVs.

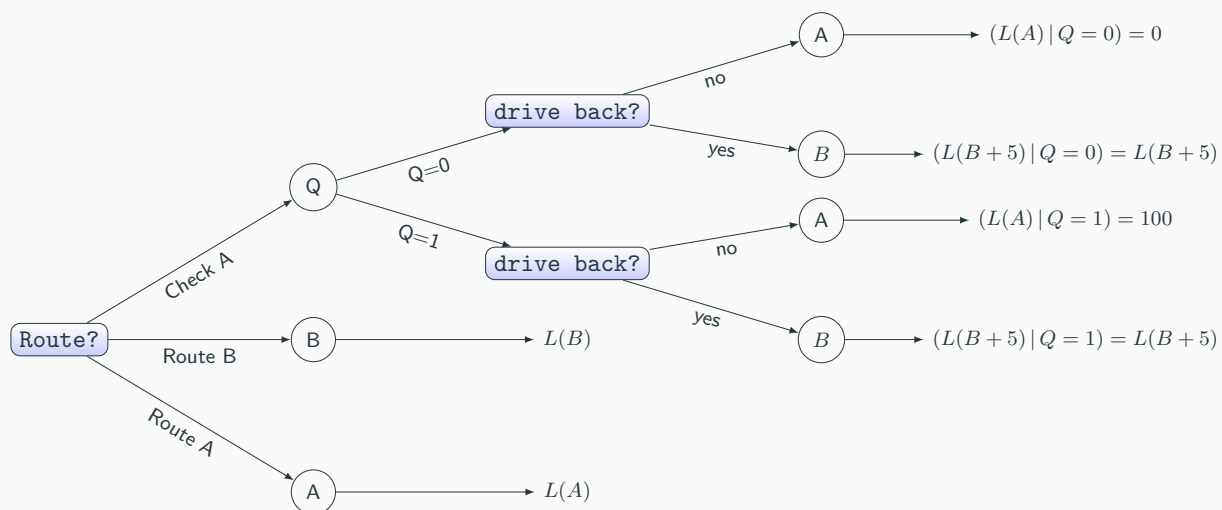
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Simple process:

- Start at leaf nodes and move towards root.
- RV nodes: Calculate expectation over utility values coming in.
- Decision nodes: Choose action corresponding to child with maximum utility. Pass on this utility.

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Example: Commuting

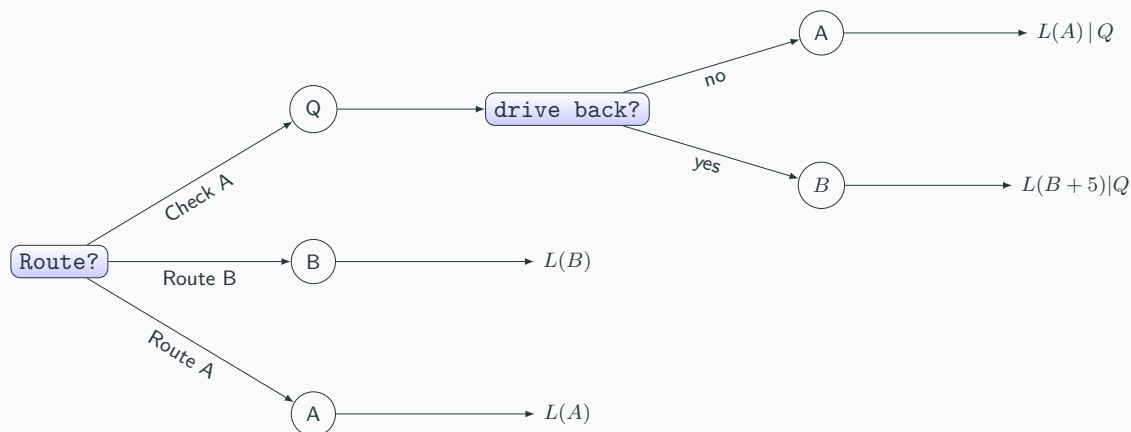


- We can split up outcomes of RVs explicitly
- This allows us to find explicit numerical values at all points.
- Note: $(L(B + 5) | Q = 0) = L(B + 5)$, since $B \perp\!\!\!\perp Q$.

Work in class.

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Example: Commuting



- Can make graph smaller by not writing out all outcomes of RVs.
- Decisions become *functions* of parent outcomes.

Work in class.

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Exploration-Exploitation Trade-Off

- We saw that “Check A” was the best option.
- Spending effort to gather information, can help us to make better decisions later.
- Uncertainty determines how much exploration can gain.
- (Would exploration be worth it if $P(Q = 1) = 0.99$?)
- → “*Exploration-exploitation trade-off.*”

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Applications and Applicability

Hugely influential theory:

- Philosophy: Utilitarianism effective altruism
- Psychology: Rational behaviour as a model for human behaviour.
- Economics: How to optimise investments.
- Game theory: Analyse implications of rational choice.
- Politics: Voting systems (Arrow's impossibility theorems).

Discussions about desirability/realism of axioms:

- Naïve application leads to bad behaviour (St Petersburg Paradox)
- Can you express desires as utility functions?
- The Law of Perverse Optimization: Whenever a desired behaviour is formulated as a utility, optimising for the utility will give you behaviour you didn't want. (AI Paperclip factory)
- Human's don't behave according to the axioms. But perhaps for good reason?

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Conclusion

- Decision theory is easy (in principle): See slide 5.
- The implications of decision theory are vast, but sadly we don't have time for this.
- Can deal with sequential nature using decision trees.
- Exploration-exploitation trade-off.

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Questions?

References:

MacKay. *Information Theory, Inference, and Learning Algorithms*, chapter 36.

Russell. *Artificial intelligence: a modern approach*, 2010, chapter 16–17.